

the familiar rule that
does NOT hold here.

$$0 \times A = 0$$



Ex. If $A = \left[\begin{array}{|c|c|} \hline \square & \boxed{\square\square} \\ \hline \end{array} \right]$ and $B = \left[\begin{array}{|c|c|} \hline \boxed{\square} & \begin{array}{|c|} \hline \square \\ \hline \square\square \end{array} \\ \hline \end{array} \right]$

then

$$A \times B =$$

$$\begin{array}{ccc} \square + \boxed{\square} & \square + \square & \square + \boxed{\square\square} \\ \boxed{\square\square} + \boxed{\square} & \boxed{\square\square} + \square & \boxed{\square\square} + \boxed{\square\square} \end{array}$$